

FIN ST1632–ST1721

Lindblad Sensitivity, Kernel Parameter Boxes,
KKT Geometry, Symmetric Decisions and the q -Clock No-Go

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Six research rounds — 90 programmes

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This report extends the conditional FIN discriminator from propagator perturbations to the declared dephasing Lindbladian, propagates explicit kernel parameter boxes, isolates the reduced continuous-minimax KKT candidate, constructs a symmetric finite-grid decision with abstention, certifies a vertex-detector matrix ball, and proves that fine-fiber q and clock scale cannot be identified separately without an anchor.

Abstract

ST1632–ST1721 advance the conditional Operational/State programme at the generator, parameter, inference and calibration levels.

For the declared energy-dephasing generator

$$\mathcal{L}_A = -i \operatorname{ad}_A - \frac{\gamma}{2} \operatorname{ad}_A^2,$$

self-adjoint graph perturbations $\|B - A\| \leq \epsilon$ satisfy

$$\|\mathcal{L}_B - \mathcal{L}_A\| \leq 2\epsilon + 2\gamma\epsilon(\|A\| + \|B\|).$$

Each semigroup is Hilbert–Schmidt contractive, so Duhamel’s formula propagates this bound to return probabilities. For $\gamma \leq 100$ and $\epsilon = 8.0418727748 \times 10^{-8}$, the certified four-time residual lower bound remains 0.10789050485132269. The theorem is specific to the jump operator A and does not cover arbitrary GKSL operators.

The strict kernel parameter map

$$w_d = \frac{\cos(\omega d + \phi)}{1 + \beta d^\eta}$$

is propagated by outward intervals. A diagnostic large box with radii $(5 \times 10^{-6}, 5 \times 10^{-6}, 5 \times 10^{-5}, 5 \times 10^{-5})$ gives $\|\Delta A\| \leq 2.0105649071 \times 10^{-4}$: it preserves the simple $t = 0.6$ channel gap but fails the existing optimum/composite sufficient gates. A small box with radii $(2 \times 10^{-9}, 2 \times 10^{-9}, 2 \times 10^{-8}, 2 \times 10^{-8})$ gives $\|\Delta A\| \leq 8.0418727748 \times 10^{-8}$ and passes. Neither box is a measured or strict confidence interval.

The reduced continuous minimax KKT geometry is isolated strongly numerically. The candidate lies on the clock boundary $\alpha = 1.1$, with

$$\gamma_* \approx 16.0591725623, qquad w \approx (0.3185956783, 0.6814043217),$$

and value 0.053705210722682, inside the certified continuous bracket $[0.05, 0.05370588868619479]$. Equal endpoint divergences and the nuisance stationarity equation are satisfied numerically; exact KKT isolation and a global active-set theorem remain open.

For the finite nuisance grid, a symmetric decision with abstention is proved. The mixture-likelihood boundary rejects the classical model, while the condition that every component likelihood is below 0.01 rejects the entire grid in favour of the classical model. Either wrong boundary has probability at most one percent under its declared null, and the boundaries are mutually exclusive. Off-grid quantum models are not covered.

An arbitrary column-stochastic vertex detector with at most five percent misclassification per true vertex changes any distribution by at most five percent in total variation. Independent classical/quantum detectors therefore leave simple-model TV at least 0.3112182255.

Finally, physical-time fine-fiber returns depend only on

$$\kappa = \frac{q}{\tau_*}.$$

The transformation $(q, \tau_*) \mapsto (cq, c\tau_*)$ leaves every number of heat and unitary fiber observations invariant. Thus q and the clock scale are not jointly identifiable without an independent anchor. Protocol 11.02 freezes these conclusions; no apparatus or physical calibration is supplied.

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1 Round I: ST1632–ST1646 — Lindblad sensitivity

On Hilbert–Schmidt operator space define

$$\text{ad}_A(X) = AX - XA, \text{quad}\mathcal{L}_A = -i\text{ad}_A - \frac{\gamma}{2}\text{ad}_A^2.$$

For $E = B - A$,

$$\|\text{ad}_B - \text{ad}_A\| \leq 2\|E\| = 2\epsilon.$$

Using

$$X^2 - Y^2 = (X - Y)X + Y(X - Y)$$

and $\|\text{ad}_A\| \leq 2\|A\|$ gives

$$\|\text{ad}_B^2 - \text{ad}_A^2\| \leq 4\epsilon(\|A\| + \|B\|),$$

and hence

$$\Delta_{\mathcal{L}equiv}\|\mathcal{L}_B - \mathcal{L}_A\| \leq 2\epsilon + 2\gamma\epsilon(\|A\| + \|B\|).$$

For self-adjoint A , $\mathcal{L}_A$ is a normal dissipative function of ad_A ; $e^{t\mathcal{L}_A}$ is Hilbert–Schmidt contractive. Thus

$$\|e^{t\mathcal{L}_B} - e^{t\mathcal{L}_A}\| \leq t\Delta_{\mathcal{L}}.$$

Rank-one state/effect Hilbert–Schmidt norms are one, so the same upper bound controls the measured return.

With $\|A\| = 2.3421820411463$, $\gamma = 100$, $t \leq 2$ and $\epsilon = 8.041872774817993 \times 10^{-8}$, the dephased-quantum plus heat- target perturbation is at most $1.5116675487625433 \times 10^{-4}$. Subtracting it from 0.10804167160619894 yields

$$0.10789050485132269 > 0.$$

Theorem 1 (declared dephasing robustness). *The four-time composite separation remains positive under the stated small noncirculant graph perturbation for every $0 \leq \gamma \leq 100$.*

This theorem does not cover independent or non- A jump operators, trace-norm contraction claims, or non-Markov heat.

2 Round II: ST1647–ST1661 — kernel parameter boxes

The nominal parameter tuple is

$$(\omega, \phi, \beta, \eta) = (0.18575, 0.16250, 1, 1.8).$$

Outward interval evaluation of all six distances constructs weight intervals, the row-sum diagonal and a circulant row-norm bound for ΔA .

Large diagnostic box. Radii

$$(5 \times 10^{-6}, 5 \times 10^{-6}, 5 \times 10^{-5}, 5 \times 10^{-5})$$

give

$$\|\Delta A\| \leq 0.00020105649070913632.$$

Every weight stays positive, and the simple return gap remains above 0.41085 by Duhamel. However this box is wider than the global-location threshold and the present $\gamma = 100$ composite sufficient bound. This is failure to certify, not a counterexample.

Small certified box. Radii

$$(2 \times 10^{-9}, 2 \times 10^{-9}, 2 \times 10^{-8}, 2 \times 10^{-8})$$

give

$$\|\Delta A\| \leq 8.041872774817993 \times 10^{-8},$$

which passes both the location and declared Lindblad-composite gates.

The interval map is conservative because the same parameters occur at every distance and in the diagonal, while elementary interval evaluation loses some correlation. More importantly, neither box has repository or experimental confidence-level provenance. The small box is a mathematical witness, not a physical error bar.

3 Round III: ST1662–ST1676 — reduced KKT candidate

The continuous minimax certificate from Release 11.01 bounds the value but not the optimizing weights. The numerical LP suggests a two-time face at $t_1 = 0.3$, $t_2 = 2.0$ and clock boundary $\alpha = 1.1$.

Near the candidate, the active divergence at each time is $D(C_i \| Q_i)$. An interior allocation requires equality of the two active endpoint divergences:

$$d_{0.3}(\gamma) = d_{2.0}(\gamma).$$

Brent isolation gives

$$\gamma_* \approx 16.05917256231611.$$

The nuisance-stationarity condition

$$w d'_{0.3}(\gamma_*) + (1 - w) d'_{2.0}(\gamma_*) = 0$$

then gives

$$w \approx 0.31859567830451857, \text{quad}1 - w \approx 0.6814043216954815.$$

The common active value is

$$0.053705210722682,$$

consistent with the certified global bracket.

The one-sided derivative in α is consistent with a minimum at the upper clock boundary, and middle times have positive reduced costs in the fine LP. These are strong numerical KKT checks. No interval Jacobian, exact root or global active-set theorem is claimed.

4 Round IV: ST1677–ST1691 — symmetric grid decision

Let $L_{j,n}$ be the likelihood ratio for grid component j , and let

$$E_n = \sum_j \pi_j L_{j,n}.$$

The decision set is

$$\begin{aligned} E_n \geq 100 &\Rightarrow Q_{\text{mixture}}, \\ \max_j L_{j,n} \leq 0.01 &\Rightarrow C_{\text{all-grid}}, \\ \text{otherwise} &\Rightarrow \text{continue/abstain}. \end{aligned}$$

Under C , E_n is an e-process, so a false quantum boundary has probability at most 0.01. If grid component j_0 is true, the event $\max_j L_{j,n} \leq 0.01$ implies $1/L_{j_0,n} \geq 100$; Ville's inequality bounds a false classical boundary by 0.01. The boundaries are mutually exclusive: if every $L_j \leq 0.01$, their convex mixture cannot exceed 0.01.

Theorem 2 (finite-grid symmetric decision). *The three-decision rule has one-percent anytime wrong-boundary control under the classical model and under every frozen grid quantum model, with abstention allowed.*

Synthetic fixtures reached the classical and quantum boundaries after 142 and a model-dependent number of events. They test code only. Off-grid quantum models have no uniform classical-boundary guarantee. A formal continuum rule would replace the maximum by $\sup_{\theta} L_{\theta,n}$ and the mixture by an exact integral; its executable certification remains open.

5 Round V: ST1692–ST1706 — detector matrix and q -clock

5.1 Vertex detector ball

Let M be column stochastic and suppose every true vertex is reported correctly with probability at least $1 - \delta$. Coupling each column with the identity gives

$$\text{TV}(Mp, p) \leq \delta.$$

For independent detector matrices,

$$\text{TV}(M_C p, M_Q q) \geq \text{TV}(p, q) - 2\delta.$$

At $t = 0.6$, the ideal full-distribution TV is 0.4112182254653598. For $\delta = 0.05$,

$$\text{TV}_{\text{observed}} \geq 0.3112182254653598.$$

The sufficient identifiability threshold is $\delta < 0.2056091127326799$. A common matrix may admit sharper Dobrushin or inverse-calibration bounds, but the triangle bound remains valid.

Loss can be represented as a thirteenth no-click outcome, restoring a column-stochastic augmented detector. Discarding it reintroduces postselection blindness.

5.2 Joint q -clock obstruction

With physical time, the fine returns are

$$H(t_{\text{phys}}) = \frac{1 + \exp[-2q t_{\text{phys}}/\tau_*]}{2}, \quad R(t_{\text{phys}}) = \cos^2(q t_{\text{phys}}/\tau_*).$$

Every record depends only on

$$\kappa = \frac{q}{\tau_*}.$$

Theorem 3 (joint q -clock no-go). *The transformation*

$$(q, \tau_*) \mapsto (cq, c\tau_*), \quad c > 0,$$

leaves all fine heat and unitary return probabilities invariant at every physical time. No number of such observations identifies q and τ_ separately.*

An independent calibration of τ_* restores the earlier conditional q -identifiability results. Multiple scheduled times alone do not.

6 Round VI: ST1707–ST1721 — synthesis

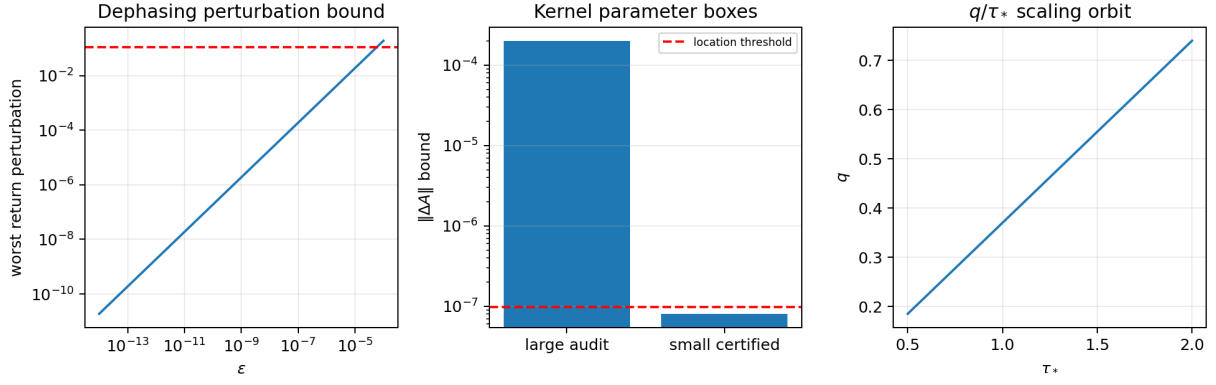


Figure 1: Left: declared dephasing perturbation bound. Centre: large and small kernel-parameter boxes relative to the location threshold. Right: the exact q/τ_* scaling orbit.

Result	Status	Boundary
declared Lindblad sensitivity	[Proven]	jump operator A , HS norm
large parameter box	[Partial]	base gap only
small parameter box	[Interval-proven]	no physical CI provenance
continuous minimax value	[Certified bracket]	exact weights open
reduced KKT geometry	[Strong numerical]	active-set theorem open
symmetric grid decision	[Proven]	finite grid, abstention
vertex detector TV ball	[Proven]	column-error class
joint q -clock identification	[Refuted]	exact scaling torsor
platform/evidence	[Absent]	no calibrated device/events

Protocol FIN_ST1707_ST1721_Protocol_11_02.json hashes Release 11.01, the continuous-prior object, independent calibration and the detector/ q -clock artifact, and records the Lindblad bound, kernel boxes, KKT candidate and symmetric decision.

These results strengthen a conditional OA only. They do not derive the state category, parameter uncertainty, clock anchor or apparatus from strict FIN.

The tests concern internal channel and parameter inference, not literal annihilation of the whole nadsoliton.

7 Recommended next programmes

1. certify a general GKSL jump-operator ball;
2. derive a parameter interval from strict kernel provenance;
3. isolate the KKT root with interval Newton/Krawczyk methods;
4. certify the continuum supremum-likelihood boundary;
5. combine detector confusion and loss in one augmented matrix polytope;
6. add a second observable with a different q/τ scaling weight;
7. impose multi-fiber q consistency;
8. optimize a continuous prior for power while preserving validity;
9. derive a platform-specific forward model;

10. freeze external clock calibration;
11. obtain independent audit;
12. collect raw events only after every freeze.

Stop if the A -dephasing bound is used for arbitrary jumps, the small box is called a measured confidence interval, the KKT candidate is called the exact optimizer, the finite grid is called a continuum decision, the TV ball is called a calibrated apparatus, or κ is called separate knowledge of q and τ_* .

8 Final conclusion

The conditional FIN discriminator now has theorem-level control at the declared Lindblad-generator and detector-matrix levels, a certified continuous minimax value, and symmetric finite-grid decisions with abstention. The fine-fiber analysis simultaneously reveals an exact scale torsor: internal returns identify only q/τ_* . The next decisive bridge is therefore not more curve fitting, but an independently sourced clock/scale observable and a platform-calibrated operator family.

A Six-round ledger

Range	Object	Verdict
ST1632–ST1646	dephasing Lindbladian	noncirculant sensitivity bound
ST1647–ST1661	kernel parameter boxes	large partial / small certified
ST1662–ST1676	reduced KKT system	strong numerical candidate
ST1677–ST1691	symmetric finite-grid decision	two boundaries plus abstention
ST1692–ST1706	detector and q -clock	TV robustness and exact scale no-go
ST1707–ST1721	synthesis	generator-to-inference robustness lattice

B Reproducibility

The authoritative result sets run from `FIN_ST1632_ST1646_Results.json` through `FIN_ST1707_ST1721_Results.json`. Each contains fifteen programmes and packet SHA-256 references. The combined test `test_fin_st1632_st1721.py` verifies all ninety packets, Lindblad bounds, parameter gates, KKT status, symmetric decision, detector TV, q -clock obstruction, protocol, figure and nonpromotion gates.

C Nonclaims

No arbitrary GKSL uncertainty theorem, physical parameter confidence interval, exact minimax optimizer, continuum symmetric decision, calibrated detector, separate q/τ_* source, state ontology, platform, raw event, independent custody, legacy bridge, selector, Standard Model, gravity, L_{total} or Theory of Everything is established. All future FIN PDFs remain English.

References

- [1] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems*, Oxford University Press.

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